

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Biology

- 1 In an agricultural region, soil was repeatedly treated with a chemical nematicide over a period of ten years to control pest worms. During this time, the population of a native worm, *Species Y*, declined continuously and eventually died out completely. In contrast, an invasive worm, *Species X*, suffered an initial decline in numbers but subsequently recovered and thrived despite continued application of the nematicide.

A student made the following statements to explain these observations:

1 The application of the nematicide acted as a selective pressure that increased the frequency of resistance alleles in the population of *Species X*.

2 The nematicide induced directed mutations in *Species X* to allow individuals to adapt to the chemical.

3 *Species Y* became locally extinct because none of its individuals possessed alleles conferring sufficient tolerance to survive the nematicide.

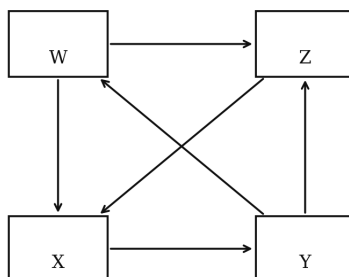
Which of the student's statements is/are correct?

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 2 The diagram shows part of the nitrogen cycle in an agricultural ecosystem containing legume crops.

	Ammonium ions in soil	Atmospheric nitrogen gas	Nitrate ions in soil	Nitrogen in plant biomass
A	W	X	Z	Y
B	X	W	Y	Z
C	X	Y	W	Z
D	Y	W	X	Z
E	Z	W	X	Y
F	X	Z	Y	W
G	Y	X	W	Z
H	Z	X	Y	W

Which row correctly identifies boxes W, X, Y, and Z?



- 3 The table shows three ecological characteristics for six freshwater fish species (A–F) inhabiting a river system subject to severe seasonal drought and habitat fragmentation:

species	dietary niche	dispersal ability	generation time
A	generalist	high	short
B	specialist	low	short
C	specialist	high	long
D	specialist	low	long
E	generalist	low	long
F	generalist	high	long

Which species is most vulnerable to local extinction as environmental conditions become persistently harsher?

- 4 A student analysed a section of a gene identified in four different organisms. The gene codes for a functional polypeptide involved in ATP synthesis.

A 15-base-pair section of the coding (non-template) DNA strand is shown in the table below. The remainder of the gene (not shown) is identical in sequence in all four organisms.

Organism	DNA sequence (5' to 3')
human	ATG CTT GGA GTC AAA
baker's yeast	ATG CTT GGT GTC AAA
pea plant	ATG CTC GGT GTT AAA
cyanobacterium	ATG AAC CGT TCG AAA

The student made the following conclusions:

1 This gene could code directly for a polypeptide subunit of ATP synthase, but cannot code directly for ATP.

2 In the cyanobacterium, transcription of this gene and translation of its mRNA can occur simultaneously.

3 Based on the section shown, the DNA sequence of the pea plant is more similar to that of the cyanobacterium than to that of baker's yeast.

Which of these conclusions is/are correct?

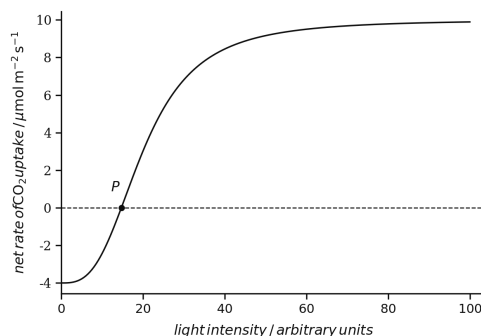
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5 A population of fish in a lake consists of two distinct phenotypes, Morph A and Morph B. Their characteristics are summarized below:
- | Trait | Morph A | Morph B |
|----------------------------|--------------------------|---------------------------------|
| Appearance | Cryptic (camouflaged) | Conspicuous (brightly coloured) |
| Competitive Ability | Weak competitor for food | Strong competitor for food |
| Metabolic Rate | Low | Low |
- Historically, the lake water was clear, and the fish were preyed upon by a visual predator. Recently, agricultural runoff has caused the water to become permanently turbid (cloudy), significantly reducing visibility. Simultaneously, the turbidity has reduced light penetration, causing a 50% decrease in the aquatic plants that serve as the food source for the fish. Which prediction for the long-term evolutionary trajectory of this population is most supported by the information provided?
- A** The relative frequency of Morph B will increase because the penalty for being conspicuous is reduced while the advantage of being a strong competitor becomes more critical.
 - B** The relative frequency of Morph A will increase because camouflage becomes even more effective in turbid water, enhancing survival.
 - C** The relative frequency of Morph A will increase because the reduction in food availability will disadvantage the conspicuous Morph B more than Morph A.
 - D** The relative frequency of Morph B will decrease because the predator will switch to non-visual hunting methods, negating the effect of turbidity.
 - E** The relative frequencies will remain unchanged because the reduction in predation pressure cancels out the reduction in food availability.

- 6 The graph shows the effect of light intensity on the net rate of carbon dioxide (CO_2) uptake by a leaf at a constant temperature and atmospheric CO_2 concentration.

Which of the following statements is/are correct?

- 1 At a light intensity of 0 arbitrary units, the rate of CO_2 release by cellular respiration is $4 \mu\text{mol m}^{-2} \text{s}^{-1}$.
- 2 At point P, the rate of the light-dependent reactions of photosynthesis is zero.
- 3 Assuming the rate of cellular respiration remains constant across all light intensities, the rate of gross CO_2 fixation at a light intensity of 80 arbitrary units is $14 \mu\text{mol m}^{-2} \text{s}^{-1}$.

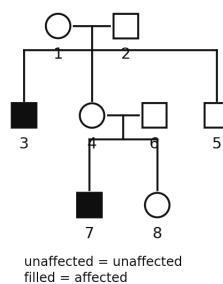


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 7 The pedigree diagram shows the inheritance of a genetic condition in a family. The condition is controlled by a single gene with two alleles. Assume complete penetrance and that no new mutations occur.

Which of the following statements is/are correct?

- 1 The condition cannot be caused by a dominant allele.
- 2 If the condition is sex-linked (X-linked) recessive, the probability that individual 8 is a carrier of the allele is 50%.
- 3 If the condition is autosomal recessive, the probability that individual 4 is a carrier of the allele is $\frac{2}{3}$.

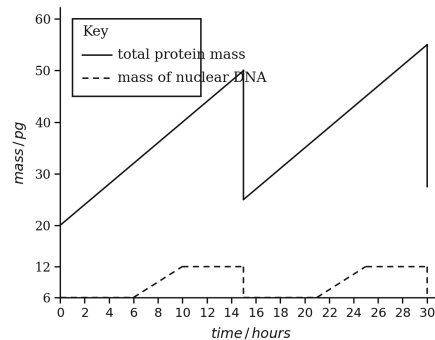


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 8 The graph shows the changes in the mass of nuclear DNA and the total protein mass of an individual dividing animal cell over a period of 30 hours.

Which of the following statements is/are correct?

- 1 Between 10 and 14 hours, the cell contains twice as many chromatids as it does between 0 and 6 hours.
- 2 The duration of the S phase is greater than the duration of the G_1 phase.
- 3 If an inhibitor of mitotic spindle formation is added to the cell at 8 hours, the mass of nuclear DNA in the cell at 18 hours will be 12.0 pg.

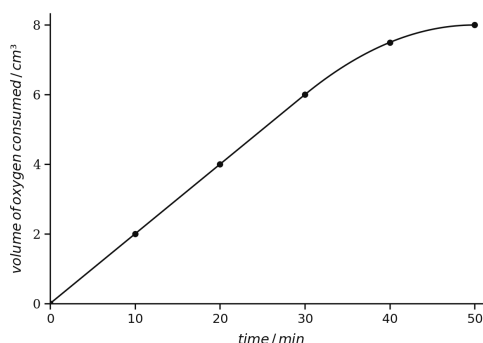


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A simple respirometer containing potassium hydroxide solution (KOH) was used to measure the aerobic respiration of germinating seeds in a sealed chamber at 20°C . The cumulative volume of oxygen consumed was recorded over a period of 50 minutes, and the results are plotted on the graph.

Which of the following statements is/are correct?

- 1 The initial rate of oxygen consumption by the seeds was $12.0\text{ cm}^3\text{ h}^{-1}$.
- 2 If the experiment were repeated with seeds metabolising only lipids ($\text{RQ} = 0.7$) without KOH , the total volume of gas in the respirometer chamber would increase over time.
- 3 The reduction in the rate of oxygen consumption after 30 minutes could be due to oxygen concentration becoming a limiting factor.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

10 In birds, sex is determined by the ZW sex-chromosome system:

- * males are homogametic (ZZ)
- * females are heterogametic (ZW)

The W chromosome does not carry an allele for the feather pattern gene.

Feather barring in chickens is controlled by a gene on the Z chromosome:

- * the dominant allele, Z^B , produces barred feathers
- * the recessive allele, Z^b , produces non-barred feathers

A breeder carries out two separate crosses:

- * **Cross 1:** a heterozygous barred male ($Z^B Z^b$) is mated with a non-barred female ($Z^b W$)
- * **Cross 2:** a non-barred male ($Z^b Z^b$) is mated with a barred female ($Z^B W$)

What is the expected proportion of all offspring that are non-barred females in Cross 1, and what is the expected proportion of all offspring that are non-barred females in Cross 2?

	Cross 1	Cross 2
A	0.25	0.00
B	0.25	0.25
C	0.25	0.50
D	0.50	0.25
E	0.50	0.50
F	0.50	1.00

- 11 Four different cell types (**P**, **Q**, **R**, and **S**) were analysed for the presence (+) or absence (–) of five cellular structures. The results are shown in the table below:

Cell type	Plasma membrane	Nucleus	Mitochondria	Ribosomes	Cell wall
P	+	+	+	+	+
Q	+	–	–	+	+
R	+	+	+	+	–
S	+	–	–	–	–

Which of the following statements is/are correct?

- 1 Cell type **P** could be a fungal cell or a plant leaf cell.
 - 2 Cell type **Q** cannot synthesise proteins or generate ATP by cellular respiration because it lacks membrane-bound organelles.
 - 3 Cell type **S** could be a mature human red blood cell (erythrocyte).
- A** 1 only
B 2 only
C 3 only
D 1 and 2 only
E 1 and 3 only
F 2 and 3 only
G 1, 2 and 3
H none of them

- 12** Haemophilia A is an X-linked recessive genetic condition in humans caused by a mutation in the gene for clotting Factor VIII. Individuals with this condition cannot produce functional Factor VIII protein, which is normally synthesised by liver cells and secreted into the bloodstream.

In a clinical trial, adult male patients with Haemophilia A receive a gene therapy treatment. A harmless viral vector carrying a functional human Factor VIII gene is injected into the bloodstream. The vector delivers the gene into the nuclei of the patient's liver cells, where it is expressed.

Which of the following statements correctly describe(s) the consequences of this gene therapy in a successfully treated patient?

- 1 The patient's liver cells are able to transcribe and translate the functional Factor VIII gene.
- 2 The patient cannot pass the mutant Factor VIII allele on to any of his future biological daughters.
- 3 The total nucleotide sequence of the DNA inside the patient's liver cells has been altered.

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 13** A student investigated stomatal density on the lower epidermis of a dicotyledonous leaf.

A transparent peel of the lower epidermis was prepared and examined under a light microscope. The circular field of view of the microscope at high magnification had an area of 0.20 mm^2 . The student counted the number of stomata in six randomly selected fields of view:

Field of view	Number of stomata
1	28
2	34
3	31
4	29
5	35
6	35

The total surface area of the lower surface of the leaf was measured to be 15.0 cm^2 .

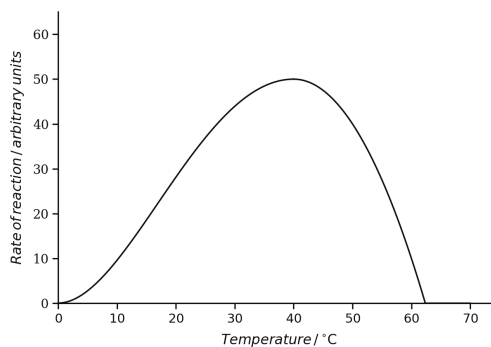
What is the estimated total number of stomata on the lower surface of this leaf?

- A** 2.4×10^3
- B** 2.4×10^4
- C** 1.2×10^5
- D** 2.4×10^5
- E** 4.8×10^5
- F** 1.44×10^6
- G** 2.4×10^6

- 14 The graph shows the effect of temperature on the rate of an enzyme-catalysed reaction.

	Percentage decrease in rate between 40 °C and 60 °C	Explanation for the decrease in rate
A	20%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
B	20%	High temperature kills the enzyme molecules, stopping their biological activity
C	40%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
D	80%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
E	80%	High temperature kills the enzyme molecules, stopping their biological activity
F	80%	The activation energy required for the uncatalysed reaction increases at high temperatures
G	400%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
H	400%	High temperature kills the enzyme molecules, stopping their biological activity

Which row correctly calculates the percentage decrease in the rate of reaction between 40°C and 60°C and gives the correct biological explanation for this decrease?



- 15** Ecologists surveyed two distinct habitats, Zone 1 and Zone 2, to compare the distribution of two plant species, X and Y. In each zone, 20 randomly placed quadrats were surveyed. The table below shows the number of quadrats in which each species was present and the total number of individuals counted for that species in that zone.

Zone	Species	Number of quadrats containing species	Total number of individuals
1	X	16	32
1	Y	5	120
2	X	8	160
2	Y	18	45

Which of the following conclusions is supported by the data?

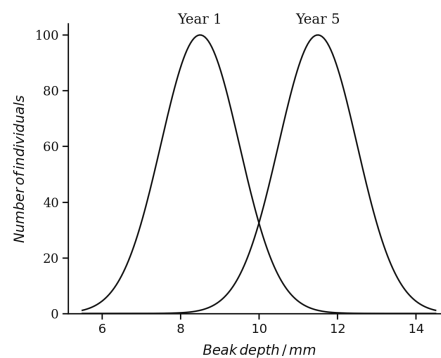
- A** In Zone 1, Species X has a higher frequency of occurrence than Species Y, despite having a lower total abundance.
- B** In Zone 2, Species X has a higher frequency of occurrence than Species Y.
- C** The frequency of occurrence of Species X in Zone 1 is 40%.
- D** Species Y demonstrates a more clumped distribution in Zone 2 than in Zone 1.
- E** The mean density of Species Y in Zone 1 is 0.25 individuals per quadrat.

- 16** A population of a seed-eating bird species lives on an isolated island. A prolonged environmental change resulted in a severe shortage of small, soft seeds, leaving predominantly large, tough-shelled seeds.

The graph shows the distribution of beak depth in the bird population in Year 1 (before the environmental change) and in Year 5 (after four generations under the new conditions).

Which of the following statements is/are correct?

- 1 The environmental change stimulated specific mutations in the birds' DNA to increase their beak depth.
- 2 The change in the population's beak depth distribution is an example of directional selection.
- 3 For this evolutionary change to occur across generations, variation in beak depth must have a genetic (heritable) basis.



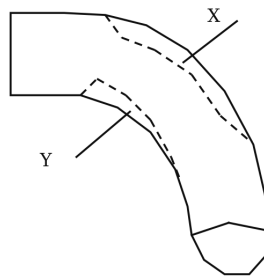
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 17** The diagram shows the tip of a plant root that was placed horizontally in the dark, with two regions in the elongation zone labelled X and Y.

A student proposed the following statements to explain the growth and bending of this root:

1. Cells in region X are longer than cells in region Y.
2. The concentration of auxin is higher in region Y than in region X.
3. The downward bending occurs because the accumulation of auxin at Y inhibits cell elongation.

Which of the student's statements is/are correct?



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 18** The table below shows the net rate of carbon dioxide uptake by a plant leaf measured at varying light intensities. The temperature and carbon dioxide concentration were maintained at constant levels throughout the experiment.

Light Intensity (arbitrary units)	Net CO ₂ Uptake (mg dm ⁻² h ⁻¹)
0	-2.0
10	0.0
30	5.5
60	9.8
90	10.5
120	10.5

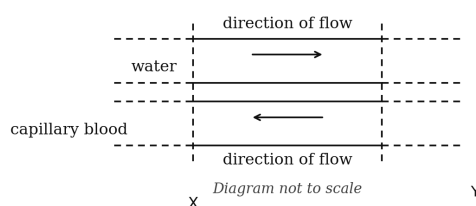
Which of the following conclusions is best supported by the data?

- A** At a light intensity of 10 arbitrary units, the rate of photosynthesis is zero.
- B** Between light intensities of 30 and 60 arbitrary units, the concentration of carbon dioxide is the primary limiting factor.
- C** The increase in net CO₂ uptake from 0 to 10 arbitrary units is primarily due to a decrease in the rate of respiration.
- D** At a light intensity of 120 arbitrary units, the rate of photosynthesis is limited by a factor other than light intensity.
- E** The maximum rate of gross photosynthesis achieved in this experiment is 10.5 mg dm⁻² h⁻¹.
- 19** The diagram shows a simplified model of gas exchange across a secondary lamella in the gill of a bony fish.

Water flows across the lamella from position X to position Y. Capillary blood flows in the opposite direction, from position Y to position X.

Which of the following statements is/are correct?

- 1 Moving along the exchange surface from position X to position Y, the partial pressure of oxygen in the capillary blood decreases.
- 2 The partial pressure of oxygen in the blood leaving at position X can exceed the partial pressure of oxygen in the water leaving at position Y.
- 3 At position Y, net diffusion of oxygen occurs from the capillary blood into the water.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** A female child is affected by a condition caused by a recessive allele of a gene located on the X chromosome.

Neither of her biological parents has the condition.

Genetic testing shows that every somatic cell in the child contains only one copy of this gene, which is the mutant allele.

Which of the following statements could explain this?

- 1 Non-disjunction of sex chromosomes occurred during meiosis in her father.
- 2 A deletion of the gene occurred on the X chromosome during meiosis in her father.
- 3 The mutant allele was inherited from an affected paternal grandfather.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** A population of insects possesses a gene with two alleles, A^* and a^* . The table below shows the probability of survival to reproductive age for each genotype in two different environments.
- | Genotype | Environment 1 | Environment 2 |
|----------|---------------|---------------|
| A^*A^* | 0.60 | 0.95 |
| A^*a^* | 0.95 | 0.95 |
| a^*a^* | 0.00 | 0.00 |
- The population has lived in Environment 1 for many generations and the frequency of allele a^* is stable. The environment then changes permanently to resemble Environment 2.

Which of the following describes the expected change in the frequency of allele a^* over the subsequent generations?

- A** It will remain constant, because the survival rate of the heterozygote Aa remains high.
- B** It will increase, because the selection pressure against the homozygous dominant genotype AA is removed.
- C** It will decrease, because the selective advantage of the heterozygote is lost while the lethal cost of aa remains.
- D** It will decrease to zero in a single generation, because individuals with the aa genotype cannot reproduce.

- 22** A continuous segment of double-stranded DNA is 300 base pairs long. In this DNA segment, 36% of the nitrogenous bases are thymine (T).

The entire DNA segment is transcribed into an mRNA molecule of 300 nucleotides. Translation of this mRNA begins at the first codon and ends at a stop codon located at the final triplet of the sequence.

Which row in the table correctly gives the total number of hydrogen bonds between the bases in this DNA segment and the number of peptide bonds formed in the resulting polypeptide?

	total number of hydrogen bonds in DNA segment	number of peptide bonds in polypeptide
A	342	98
B	342	99
C	684	98
D	684	99
E	684	100
F	816	98
G	816	99
H	816	100

- 23 The table shows some mRNA codons and the amino acids or translation signals they encode:

mRNA codon	Amino acid or function
AUG	Met (Start)
CAA	Gln
CAG	Gln
GAA	Glu
GAG	Glu
GGA	Gly
GGC	Gly
UUA	Leu
UUG	Leu
UCA	Ser
UCG	Ser
UGG	Trp
UAA	Stop
UAG	Stop
UGA	Stop

An unmutated mRNA molecule has the sequence:

5'–AUG UUA GAA GGA UCG UGG GAG CAA UAA–3'

Single nucleotide substitution mutations took place at both the 8th and 17th nucleotides in this sequence.

Using **only** the information provided, which of the following statements **could** be correct for the polypeptide produced by translating this mutated sequence from left to right?

- 1 The resulting polypeptide could be only five amino acids long.
- 2 The resulting polypeptide could contain only four different types of amino acids.
- 3 The resulting polypeptide could contain seven different types of amino acids.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 24** Researchers are investigating a homeostatic defect in a mutant strain of mice. They compare the blood glucose concentration of mutant and wild-type mice under two experimental conditions:

1. **Glucose Tolerance Test:** Mice are injected with a glucose solution.
2. **Fasting Test:** Mice are withheld from food for 12 hours.

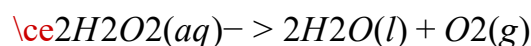
The results are summarised below:

Condition	Wild-type response	Mutant response
Glucose Tolerance Test	Blood glucose rises, then returns to baseline within 90 minutes	Blood glucose rises, then returns to baseline within 90 minutes
Fasting Test	Blood glucose remains stable within the normal range	Blood glucose decreases rapidly to hypoglycaemic levels

Which of the following defects best explains these results?

- A** The mutant mice are unable to synthesise insulin.
- B** The mutant mice have insulin receptors that are permanently activated.
- C** The mutant mice constitutively secrete high levels of insulin.
- D** The mutant mice are unable to secrete glucagon.
- E** The mutant mice have glucagon receptors that are permanently activated.

- 25** A student investigated the rate of an enzyme-catalysed reaction using catalase to break down hydrogen peroxide into water and oxygen gas:



Under standard control conditions, 10 cm³ of 2.0% hydrogen peroxide solution was mixed with 1.0 cm³ of catalase solution at 25 °C and pH 7.0. The volume of oxygen gas produced over time was recorded, giving curve **X** on the graph.

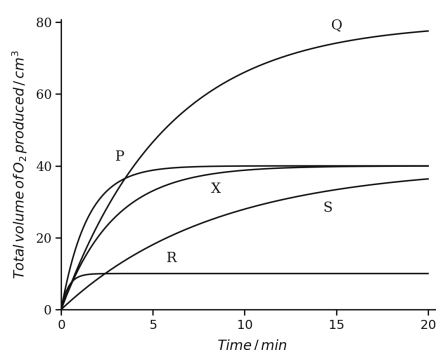
The student then carried out four separate variations of the experiment:

- **Variation 1:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 2.0 cm³ of catalase solution at 25 °C and pH 7.0.
- **Variation 2:** 20 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 25 °C and pH 7.0.
- **Variation 3:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 75 °C and pH 7.0.
- **Variation 4:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 10 °C and pH 7.0.

Assume the catalase has an optimum temperature of 37 °C and denatures rapidly above 60 °C.

Which row in the table correctly identifies the curve corresponding to each variation?

	Variation 1	Variation 2	Variation 3	Variation 4
A	P	Q	R	S
B	P	Q	S	R
C	P	S	R	Q
D	Q	P	R	S
E	Q	S	P	R
F	S	P	R	Q
G	S	Q	P	R
H	S	Q	R	P



- 26** Cylinders of potato tissue were immersed in sucrose solutions for 24 hours. The results are shown below.

Water potential of solution, Ψ / MPa	Mean change in mass / %
-0.2	+12
-0.5	+6
-0.8	0
-1.1	-6

At no net mass change, the tissue water potential equals the solution water potential. For this question, cell water potential is given by $\Psi = \Psi_s + \Psi_p$, where Ψ_s is solute potential and Ψ_p is pressure potential.

At equilibrium, $\Psi_p = 0.25$ MPa. What is Ψ_s ?

- A** -0.55 MPa
B -0.80 MPa
C -1.05 MPa
D -1.35 MPa
- 27** A unicellular organism was analysed and found to possess the following features:
- * circular double-stranded DNA not enclosed by a nuclear envelope
 - * 70S ribosomes in the cytoplasm
 - * no membrane-bound organelles
 - * a surrounding cell wall
- Which of the following statements is/are correct?
- 1 It cannot carry out aerobic respiration.
 - 2 It possesses genes that code for ribosomal RNA.
 - 3 It reproduces asexually by binary fission rather than mitosis.
- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3